Healthcare Patient & Hospital Performance Analysis

Project 4 — Multi-Domain Data Analysis Portfolio

Analysis of 2,000 simulated patient records using Python, Pandas, NumPy and Matplotlib

Purpose: Evaluate hospital operations, treatment costs, patient satisfaction and readmission patterns, and translate findings into actionable management recommendations.

Important: The dataset is simulated for educational and portfolio demonstration purposes and is not intended for clinical decision-making.

1. Executive Summary

This project analyzes 2,000 simulated healthcare patient records to understand hospital department performance, treatment costs, length of stay, patient satisfaction and readmission patterns. The average patient age was 49.66 years, average hospital stay was 7.40 days, average treatment cost was ■51,319.68, and average satisfaction was 5.43/10. The overall readmission rate was 14.8%. General Medicine recorded the highest average treatment cost, while Cardiology had the highest department-level readmission rate. At diagnosis level, Hypertension had the highest readmission rate. Correlation analysis identified a moderate positive relationship between treatment cost and cost per day and a stronger negative relationship between length of stay and cost per day. The results highlight opportunities for improving patient experience, monitoring readmissions, reviewing high-cost areas and optimizing discharge planning.

2. Project Objectives

• Analyze patient demographics and hospital departments.

• Compare treatment costs across departments and diagnoses.

• Analyze hospital length of stay and cost efficiency.

• Evaluate patient satisfaction levels.

• Measure readmission rates by department and diagnosis.

• Use descriptive statistics and correlation analysis to identify patterns.

• Provide actionable recommendations for hospital management.

3. Dataset & Methodology

The dataset contains 2,000 simulated patient records and 12 analytical fields. Derived variables include Age Group, Cost per Day and Satisfaction Level. The workflow used Pandas for data manipulation and aggregation, NumPy for data generation and numerical operations, and Matplotlib for visualization.

Metric	Result
Records	2,000
Columns	12
Missing values	0
Duplicate rows	0
Unique patient IDs	2,000
Age range	18–80 years
Length of stay	1–14 days
Treatment cost	■5,022.90–■99,924.69
Satisfaction score	1–10

4. Key Healthcare KPIs

KPI	Value
Average age	49.66 years
Average length of stay	7.40 days
Average treatment cost	■51,319.68
Median treatment cost	■50,800.46

Average satisfaction	5.43 / 10
Overall readmission rate	14.8%

5. Department Analysis

General Medicine had the highest average treatment cost at ■52,473.54. Neurology had the largest patient volume with 421 patients. Pediatrics had the longest average stay at 7.50 days. Cardiology recorded the highest department-level satisfaction at 5.63/10 and the highest readmission rate at 16.83%.

Department	Patients	Avg Cost	Avg Stay	Avg Satisfaction	Readmission
General Medicine	398	■52,473.54	7.40	5.40	15.08%
Neurology	421	■51,772.92	7.38	5.56	13.78%
Pediatrics	406	■51,705.02	7.50	5.28	12.81%
Orthopedics	365	■50,925.31	7.21	5.27	15.62%
Cardiology	410	■49,703.71	7.49	5.63	16.83%

6. Diagnosis Analysis

Diabetes had the highest patient volume with 430 patients. Infection had the highest average treatment cost at ■52,579.67. Hypertension had the longest average stay at 7.67 days and the highest diagnosis-level readmission rate at 16.45%. Migraine had the highest diagnosis-level satisfaction at 5.52/10, while Infection and Fracture had the lowest at 5.32/10.

Diagnosis	Patients	Avg Cost	Avg Stay	Avg Satisfaction	Readmission
Diabetes	430	■50,632.49	7.36	5.47	14.88%
Infection	407	■52,579.67	7.56	5.32	15.72%
Migraine	400	■51,300.92	7.05	5.52	15.25%
Hypertension	383	■49,837.99	7.67	5.54	16.45%
Fracture	380	■52,260.92	7.38	5.32	11.58%

7. Patient Satisfaction

The average satisfaction score was 5.43/10. Satisfaction was distributed across 686 low, 685 medium and 629 high satisfaction patients. The largest category was therefore Low satisfaction (34.3%), indicating an opportunity to improve communication, waiting-time management and service quality.

8. Readmission Analysis

There were 296 readmitted patients out of 2,000, producing an overall readmission rate of 14.8%. Cardiology had the highest department readmission rate at 16.83%, while Pediatrics had the lowest at 12.81%. By diagnosis, Hypertension had the highest rate at 16.45% and Fracture the lowest at 11.58%.

9. Correlation Analysis

Variable Pair	Correlation	Interpretation
Treatment Cost ↔ Cost per Day	0.425	Moderate positive relationship
Length of Stay ↔ Cost per Day	-0.614	Strong negative relationship
Age ↔ Length of Stay	0.039	Very weak relationship
Length of Stay ↔ Treatment Cost	-0.018	Very weak relationship
Satisfaction ↔ Treatment Cost	-0.002	Virtually no relationship
Satisfaction ↔ Length of Stay	0.044	Very weak relationship

Correlation measures association, not causation. The relationships above describe patterns in the simulated dataset.

10. Visual Analysis

Patient Distribution by Department — visualization file not found at report-generation time.

Average Treatment Cost by Department — visualization file not found at report-generation time.

Readmission Rate by Department — visualization file not found at report-generation time.

Average Treatment Cost by Diagnosis — visualization file not found at report-generation time.

Patient Satisfaction by Department — visualization file not found at report-generation time.

Length of Stay vs Treatment Cost — visualization file not found at report-generation time.

Patient Satisfaction Distribution — visualization file not found at report-generation time.

11. Business Recommendations

Improve patient experience

Focus on communication, waiting times, service quality and structured patient feedback to address the sizeable low-satisfaction group.

Monitor readmission risk

Prioritize departments and diagnosis groups with higher observed readmission rates, especially Cardiology and Hypertension, for closer post-discharge follow-up.

Review high-cost areas

Investigate high-cost departments and diagnoses to identify resource-efficiency opportunities without compromising care quality.

Optimize discharge planning

Review longer stays and discharge workflows to identify operational bottlenecks and improve bed utilization.

Build KPI monitoring

Track treatment cost, length of stay, satisfaction and readmission rates in a recurring healthcare dashboard.

12. Data Validation Evidence

Validation confirmed 2,000 rows, 12 columns, zero missing values, zero duplicate rows and 2,000 unique Patient IDs. Value ranges were also checked for age, length of stay, treatment cost and satisfaction score. Categorical fields contained only the expected department, diagnosis and readmission values.

13. Conclusion

This project demonstrates a complete healthcare data-analysis workflow: simulated data generation, feature creation, validation, descriptive statistics, aggregation, correlation analysis, visualization and business recommendations. The results provide a portfolio-ready example of how data analysis can support operational and patient-experience decisions. Because the dataset is simulated, the findings should be treated as analytical examples rather than real clinical evidence.